

VR Training: The Unused Opportunity to Save Lives During a Pandemic

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Figure 1: Representation of the virtual environment in which the priming of an ECMO machine can be trained.

ABSTRACT

When on life support, the patients' lives not only depend on the availability of the medical devices but also on the staff's expertise to use them. With the example of ECMO devices, which were highly demanded during the COVID-19 pandemic but rarely used until then, we developed a VR training for priming an ECMO to provide the required expertise in a standardized and simple way on a global scale. This paper presents the development of VR training with feedback from medical and technical experts.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models

1 INTRODUCTION

The global COVID-19 pandemic has demonstrated how fast medical capacity can reach its limits. A COVID-19 infection can damage the lungs and lead to acute lung failure. The last therapeutic option for acute lung failure is extracorporeal membrane oxygenation (ECMO) [5]. When the lungs can no longer provide sufficient oxygen to the blood, ECMO fills the blood with oxygen outside the body. However, with such therapy, the survival of patients depends on the medical staff's experience. Barbaro *et al.* have indicated that hospitals in which more than 30 adult patients annually receive an ECMO treatment have significantly lower mortality odds than hospitals in which less than six annual cases are treated [2]. Besides, Banjas *et al.* reported that it takes about three years and 50 treatments for a hospital department to gain the necessary experience for successful ECMO treatment [1]. Furthermore, considering the aggravated conditions of the COVID-19 pandemic (*e.g.*, number of

therapies, lack of time, resources, and personnel), it can be inferred that with the required expertise the mortality rate of ECMO patients would have been even lower [3].

Therefore seeing how critical training is to the treatment outcome, we have developed a successful virtual reality (VR) training application, as shown in Figure 1. It allows medical staff to learn how to prime an ECMO machine. This training offers the advantages that, on the one hand, the training with such technology can be better than with conventional training [6]. On the other hand, such an application can be deployed worldwide and be available within a very short time. In the following, we describe the development of the application and the resulting findings. We intend to present the possibilities and problems of such training since this technology can be used to support medical personnel and save patients' lives in a future pandemic.

2 ECMO: PRIMING-TRAINING

2.1 Development Concept

For the application development, we received a part of the required 3D models by the manufacturer of the ECMO machine, which we optimized for VR. The remaining objects were digitally reconstructed with the 3D creation suite Blender. Regarding the software development, we used the Unity version 2019.3.9f1. The software architecture was designed to be cross-platform (*e.g.*, SteamVR or Oculus) in order to reduce hardware-specific limitations as far as possible and to enable global distribution of the application. For this reason, we also implemented the application in multiple languages.

2.2 Training Steps

First, we defined the training content based on the manufacturer's official manual and training video, which led to 41 different training steps. An excerpt of necessary priming steps is depicted in Table 1. The virtual screen illustrated in Figure 1 is used to visualize the individual steps to the user. Since the majority of the users are not familiar with this medical device and thus may not know the technical names of certain components, animated hints highlight the currently relevant objects (*c.f.* Figure 2 (A)).

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Table 1: Excerpt of the 41 training steps in priming an ECMO: These steps are presented in more detail during the VR training.

Step	Task	Action
1	Put on the sterile gloves	Tap
2	Connect the priming set	Grab
3	Place the gas exchanger on the holder	Grab
⋮	⋮	⋮
40	Set the O-Flow by air bubbles	Tap
41	Close the menu navigation	Tap

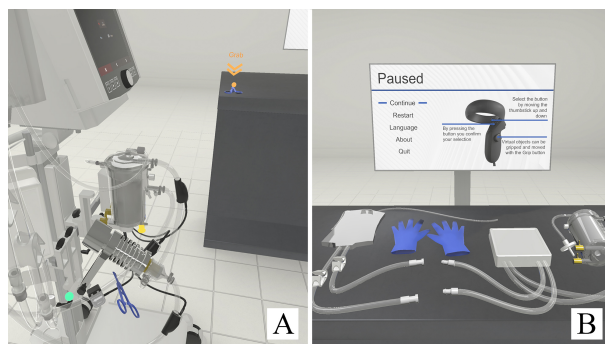


Figure 2: A: Hints for the grabbable object (orange dot) and its target position (green dot). B: Virtual monitor displaying the respective task and the interactable menu.

2.3 Training Interaction

To avoid unnecessarily overloading or distracting the users, we limited the interaction options. As shown in Table 1, the user interactions at each step of the priming procedure can be performed by tapping and grabbing objects. The tap interactions are employed when, e.g., virtually touching of a switch or a clamp. This is triggered when the target object collides with the virtual hand. Grab interactions are utilized when objects are placed at specific positions by grabbing and moving them with the controller's grab trigger. Once the object is correctly placed on the target position, it is positioned automatically, and the next step begins.

The menu is used to control the application by providing users with the functionality to select the desired language, stop, pause or restart the training. A controller button can be used to access the menu and confirm the desired actions. The respective menu items can be selected by using the controller's thumbstick. When the user enters the menu, it is visualized on the virtual screen, as shown in Figure 2 (B), and the training is paused.

3 RESULTS

Due to the high demand for available ECMO devices and the equipment shortages, we were not able to evaluate the VR training by a study yet [4]. However, it is reasonable to assume that the training success of the ECMO training is equal to or better than that of the conventional training. This can be deduced by the fact that a conceptually comparable VR training, which focused on the priming procedure of a dialysis machine, was already evaluated with convincing results [6]. Regarding learning success, this study indicated that VR training outperformed both video training and traditional group training by an instructor.

In order to obtain new findings for optimizing VR training, we provided the soft- and hardware (VR glasses) to several medical institutions and the medical device manufacturer. The resulting feed-

back was mainly positive. It was noted that the virtual environment is clearly arranged, the usability is easy after a few attempts, the respective steps are understandable as well as easy to learn, and the possibilities of the VR training are convincing. Feedback on the content included turning a pump head into a specific position instead of simply placing it at the desired position. Moreover the question was posed what the technical limitations would be regarding very detail-intensive processes.

4 CONCLUSION

So far, the developed VR application has not yet been released to medical institutions due to certain regulations regarding the instruction of medical devices. For instance, according to the regulation in Germany, there are no requirements for the type of instruction. So, a movie or software-based instruction is permissible. Nevertheless, a way for the instructed person to ask questions, e.g., by phone, must be provided. This is currently not feasible because both medical and manufacturing capacities are overloaded during a pandemic.

Overall, VR training provides a variety of possibilities and advantages that can influence the mortality rate during a pandemic. On the one hand, it has already been proven that this method's training success is more convincing than conventional training [6]. On the other hand, this technology offers the advantage that the limited resources (e.g., devices, equipment) are not wasted for training but can be used for patient care with the necessary expertise [3]. Moreover, such training can be performed worldwide, at any time and as often as required. Likewise, it is possible to measure the user's training success in the application.

Therefore, we recommend the establishment of a worldwide network of experts, which is able to bundle the relevant information and integrate them into a platform. This would allow medical professionals to train in a simple, flexible, and standardized way. It should also be noted that any knowledge like inclusion and exclusion criteria or treatment regimens can be shared and trained using this technology.

The COVID-19 pandemic has demonstrated how fast a pandemic can change all our lives. The gained experience must be used to act more effectively and efficiently, and minimize the associated consequences, not only for potential future pandemics.

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